

# Pragyan Shukla

Fairfax, VA | [LinkedIn](#) | [GitHub](#) | +1 (703) 364-8385 | [pshukl@gmu.edu](mailto:pshukl@gmu.edu) | U.S. Citizen

## EDUCATION

### George Mason University (2022-2025)

Fairfax, Virginia

*BS in Computer Science*

RISE Scholar 2024

UTA for CS 110 (2023)

### Vellore Institute of Technology (2020-2022)

Vellore, Tamil Nadu, IN

*B.Tech in Computer Science and Engineering with specialization in Business Systems*

International Transfer Student

## SKILLS & INTERESTS

**Programming & Development:** Python, Java, Git, HTML, CSS, React, AWS (S3, Lambda, SES, SNS)

**AI & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, OpenAI API, NLP

**Software & Tools:** Docker, SQL, MongoDB

**Interests:** AI, Machine Learning, Data Analytics, Software Development, Problem Solving, Reading, Running

## EXPERIENCE

### Chartway Credit Union

Virginia Beach, VA

*Data Engineering Intern*

May 2025– July 2025

- Engineered a data-observability framework for core-banking and loan platforms, using dynamic T-SQL to surface four historic data-quality events.
- Authored parameter-driven stored procedures in SQL Server (SSMS / Visual Studio Database Projects) to automate audits that once took engineers hours—now finished with a single call.
- Maintained source control with GitKraken and enforced pull-request policies in Azure DevOps, collaborating in two-week Agile/Scrum sprints (stand-ups, sprint reviews, retros).
- Preparing a capstone presentation for senior leadership, showcasing framework design, error-tracking workflow, and roadmap for real-time alerting (email/Teams).

### George Mason University - College of Science

Fairfax, VA

*Research & Interdisciplinary STEM Experiences (RISE) Scholar 2024*

Jan 2024– June 2024

- Conducted research on predicting photovoltaic (PV) power of solar panels using supervised machine learning models.
- Developed and optimized models including Support Vector Regression (SVR), Polynomial Regression, Decision Trees, and Random Forest, achieving a 15% improvement in  $R^2$  score over baseline models.
- Employed error metrics (MAE, MSE, RMSE,  $R^2$ ) to evaluate model performance and optimize predictions.
- Presented research findings at the RISE Symposium, creating a visually compelling poster and report.

### ByteRatio Inc.

Leesburg, VA

*Jr. Software Developer Intern*

May 2023– July 2023

- Designed and deployed an Excel automation application using Python and AWS, reducing processing time from 8 hours to 5 minutes.
- Engineered an optimized workflow, boosting efficiency by 95% through automation and cloud computing.
- Leveraged AWS S3 for storage, Lambda for serverless execution, SES/SNS for notifications, and Jira/Confluence for project management.
- Collaborated with a cross-functional team to develop scalable solutions improving productivity.

### The Sparks Foundation

Remote

*Data Science and Business Analyst Intern*

December 2021 – January 2022

- Developed data collection and analysis pipelines, integrating internal and external datasets for business intelligence insights.
- Applied machine learning models (Regression, Classification) using Python & R to extract valuable trends.
- Strengthened soft skills through networking workshops and professional development activities.

## PROJECTS

### AutoFind: A Semantic Search Engine for Cars

- Designed and implemented an AI-powered semantic search engine for cars using Cohere's NLP models, improving search relevance by 30%.
- Built a web scraper to auto-fill missing "Market Category" data from automotive websites, ensuring a 100% complete dataset.
- Fine-tuned large-scale NLP models to improve query understanding and response accuracy.
- Deployed a full-stack solution integrating ML models, APIs and a user-friendly low code web interface.